Want to Stop Face being Misused?

IDProtector prevents encoder-based ID-preserving generation

Image created with ChatGPT.

IDProtector: An Adversarial Noise Encoder to Protect Against ID-Preserving Image Generation

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Introduction

Goal

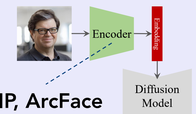
Generate with

- Wrong ID
- Bad quality

Requirements

- Fast
- Robust
- Imperceptible

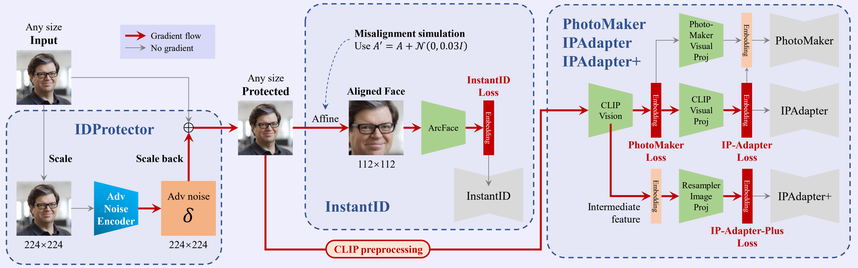
Encoder-Based Generation



CLIP, ArcFace

InstantID, PuLID, IP-Adapter, IP-Adapter-Plus, PhotoMaker, Face0, IP-Adapter-FaceID, IDAdapter, Storymaker

Methodology



We train **IDProtector** to solve the optimization problem

$$\min_{\theta} S(g(x), g(x+\delta)) \quad \delta = E_\theta(x)$$

Image Generator; Face Similarity; IDProtector Adv noise encoder

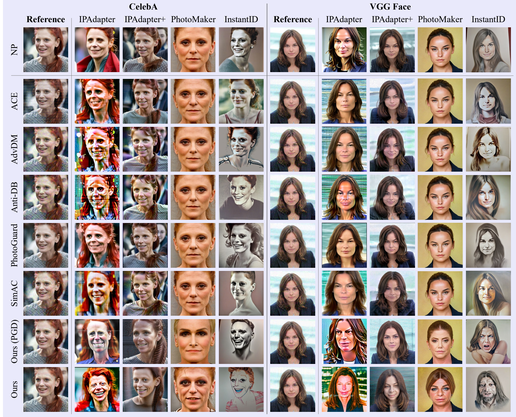
subject to the constraint

$$\|\delta\|_\infty < \epsilon$$

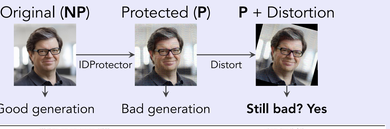
Main Experiments

Lower ISM	= **Protection is effective**
PGD good performance	= **Objective is effective**
Equal performance on VGGFace	= **Dataset agnostic**

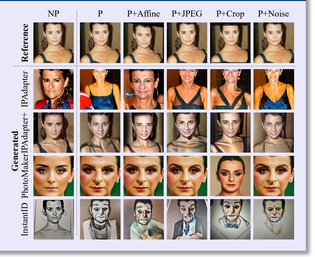
Dataset	Protection Method	IP-Adapter FDR ↓	IP-Adapter ISM ↓	IP-Adapter SER-FIQ ↓	IP-Adapter-Plus FDR ↓	IP-Adapter-Plus ISM ↓	IP-Adapter-Plus SER-FIQ ↓	PhotoMaker FDR ↓	PhotoMaker ISM ↓	PhotoMaker SER-FIQ ↓	InstantID FDR ↓	InstantID ISM ↓	InstantID SER-FIQ ↓
CelebA	No Protect	1.000	0.176	0.641	1.000	0.240	0.663	0.664	0.265	0.605	0.996	0.775	0.774
	ACE	1.000	0.159	0.640	1.000	0.231	0.681	0.636	0.268	0.594	1.000	0.745	0.758
	AdvDM	0.988	0.157	0.630	1.000	0.226	0.666	0.640	0.255	**0.590**	1.000	0.757	0.757
	Anti-DB	1.000	0.155	0.615	1.000	0.222	0.676	0.640	0.255	0.599	1.000	0.753	0.752
	PhotoGuard	1.000	0.152	0.621	1.000	0.223	0.670	0.652	0.259	0.592	1.000	0.758	0.767
	SimAC	1.000	0.162	0.627	1.000	0.217	0.683	0.652	0.253	0.599	1.000	0.752	0.755
	Ours (PGD)	**0.844**	**0.024**	**0.544**	**0.912**	**0.072**	0.718	**0.112**	**0.094**	0.639	**0.996**	**-0.166**	0.721
	Ours	0.952	0.060	0.688	0.996	0.114	**0.644**	0.672	0.123	0.622	1.000	0.231	**0.613**
VGG Face	No Protect	1.000	0.201	0.672	1.000	0.236	0.698	0.632	0.254	0.600	0.996	0.764	0.828
	ACE	0.988	0.154	0.617	1.000	0.209	0.683	1.000	0.209	0.683	1.000	0.703	0.829
	AdvDM	1.000	0.147	**0.597**	1.000	0.215	0.693	1.000	0.215	0.693	1.000	0.706	0.820
	Anti-DB	1.000	0.138	**0.597**	1.000	0.210	0.678	**0.624**	0.238	**0.599**	1.000	0.707	0.811
	PhotoGuard	1.000	0.159	0.643	1.000	0.217	0.686	**0.624**	0.229	0.612	**0.996**	0.691	0.799
	SimAC	1.000	0.150	0.605	1.000	0.202	**0.677**	**0.624**	0.228	0.644	1.000	0.707	0.826
	Ours (PGD)	**0.968**	**0.043**	0.669	1.000	0.202	0.740	**0.624**	0.228	0.630	1.000	0.707	0.769
	Ours	1.000	0.105	0.737	1.000	**0.138**	0.687	0.632	**0.126**	0.614	1.000	**0.232**	**0.713**



Experiments: Robustness



	IDProtector (ours, ViT) IP-Adapter FDR ↓	IP-Adapter ISM ↓	IP-Adapter+ FDR ↓	IP-Adapter+ ISM ↓	PhotoMaker FDR ↓	PhotoMaker ISM ↓	InstantID FDR ↓	InstantID ISM ↓	Adaptive PGD IP-Adapter FDR ↓	IP-Adapter ISM ↓	IP-Adapter+ FDR ↓	IP-Adapter+ ISM ↓	PhotoMaker FDR ↓	PhotoMaker ISM ↓	InstantID FDR ↓	InstantID ISM ↓
NP	1.000	0.176	1.000	0.240	0.664	0.265	**0.996**	0.775	0.844	0.024	**0.912**	0.072	0.112	0.094	**0.996**	-0.166
P	**0.952**	**0.060**	0.996	0.114	0.672	**0.123**	1.000	0.231	0.844	0.024	**0.912**	0.072	0.112	0.094	**0.996**	-0.166
P+Affine	1.000	0.068	1.000	0.106	0.656	0.135	1.000	**0.216**	0.844	0.064	0.976	0.081	0.112	-0.009	1.000	-0.147
P+JPEG	0.964	0.069	1.000	**0.097**	0.660	0.138	1.000	0.241	0.784	**0.015**	0.948	**0.062**	**0.108**	**-0.012**	1.000	-0.161
P+Crop	1.000	0.072	1.000	0.108	**0.652**	0.132	1.000	0.216	**0.752**	0.058	0.940	0.070	0.108	0.020	1.000	**-0.167**
P+Noisy	**0.952**	0.069	**0.984**	0.114	0.656	0.132	1.000	0.252	0.832	0.023	0.932	0.072	0.112	-0.010	1.000	-0.130
Avg (P+)	0.979	0.070	0.996	0.106	0.656	0.134	1.000	0.231	0.803	0.040	0.949	0.071	0.110	-0.003	1.000	-0.151



Experiments: Generalization

Protection generalizes to some generators unseen during training

	IPA-FIDXL	IPA-FID+	StoryMaker	Midjourney	Jing Gou
Average ISM (CelebA-test)	0.523 → 0.132	0.644 → 0.217	0.572 → 0.193	0.199 → 0.098	0.672 → 0.131

Experiments: Time & Image Quality

Faster protection, better invisibility

	ACE	AdvDM	Anti-DB	Photo-Guard	SimAC	Ours PGD	Ours
Time (s)	87.0	51.9	261	23.3	323	4089	**0.173**
PSNR (dB)	31.06	31.70	31.70	31.10	32.37	**32.94**	32.15
SSIM (dB)	0.741	0.766	0.765	0.757	0.791	**0.845**	0.809

arXiv 2412.11638